

SSC CGL TIER-1: TOPIC-WISE QUESTION PAPERS & ≤5-SECOND SHORTCUTS

GROUNDING IN OFFICIAL NOTIFICATION & HIGH-TO-LOW WEIGHTAGE MASTER BLUEPRINT

Quantitative Aptitude (17 Topics) • Reasoning (15 Topics) • English (9 Topics) • General Awareness (9 Topics)
Every Topic Equipped with: Master Formula • The ≤5-Second Shortcut • Official Shift-Level Practice Papers • Instant Optical Solutions

SECTION 1: QUANTITATIVE APTITUDE

25 Qs | 50 Marks | Target: 24-25 Correct

Topic #1: Data Interpretation (Expected: 3 – 6 Qs Qs | Priority: Very High)

Speed Target: ≤5 Sec

Master Formula: Percentage = (Value / Base) × 100 | Degree to % = (Angle / 360) × 100 = Angle × (5/18)%

≤5-Second Shortcut: Truncate 4-digit numbers to 2 digits. Pre-calculate 1° = 200 units. Match dominant pie slice ratios optically.

Q1. In a table DI: Production in 2021 = 4,182 tons; Production in 2022 = 8,391 tons. What is the approximate percentage increase?

Options: A) 95.2% B) 100.6% C) 105.4% D) 110.2%

≤5s Solution: Truncate first 2 digits: 42 → 84 is exactly +100%. Since 8391 is slightly over double, pick 100.6% in 3s. → Answer: B (100.6%)

Q2. In a Pie Chart with total 360° = 72,000 employees. What is the number of employees in Sector A with angle 54°?

Options: A) 9,600 B) 10,800 C) 12,000 D) 14,400

≤5s Solution: 54° = (54/360) = 3/20 = 15%. 15% of 72,000 = 7,200 + 3,600 = 10,800 in 2s. → Answer: B (10,800)

Topic #2: Geometry (Triangles & Circles) (Expected: 2 – 4 Qs Qs | Priority: Very High)

Speed Target: ≤5 Sec

Master Formula: DCT = $\sqrt{d^2 - (r_1 - r_2)^2}$ | Incenter $\angle BIC = 90^\circ + A/2$ | Tangent-Secant $PT^2 = PA \times PB$

≤5-Second Shortcut: Pythagorean triplets (8-15-17, 5-12-13, 7-24-25). Circumcenter = 2A, Orthocenter = 180° - A.

Q3. Two circles of radii 10 cm and 2 cm have centers 17 cm apart. What is the length of their Direct Common Tangent (DCT)?

Options: A) 12 cm B) 15 cm C) 16 cm D) 18 cm

≤5s Solution: DCT = $\sqrt{17^2 - (10-2)^2} = \sqrt{289 - 64} = \sqrt{225} = 15$ cm (Pythagorean 8-15-17 triplet in 3s). → Answer: B (15 cm)

Q4. In $\triangle ABC$, I is the incenter. If $\angle BAC = 70^\circ$, find the measure of $\angle BIC$.

Options: A) 110° B) 125° C) 140° D) 145°

≤5s Solution: $\angle BIC = 90^\circ + (70^\circ / 2) = 90^\circ + 35^\circ = 125^\circ$ in 1s. → Answer: B (125°)

Q5. Secant PAB intersects circle at A and B (PA = 4 cm, AB = 5 cm). PT is tangent from P. Length of PT = ?

Options: A) 5 cm B) 6 cm C) 7 cm D) 8 cm

≤5s Solution: PB = 4 + 5 = 9 cm. $PT^2 = PA \times PB = 4 \times 9 = 36 \rightarrow PT = 6$ cm in 2s. → Answer: B (6 cm)

Topic #3: Mensuration (2D & 3D) (Expected: 2 – 4 Qs Qs | Priority: Very High)

Speed Target: ≤5 Sec

Master Formula: Cone CSA = $\pi r l$ | Cylinder TSA = $2\pi r(r+h)$ | Sphere Vol = $(4/3)\pi r^3$ | Recast N = $(R/r)^3$

≤5-Second Shortcut: The π 11-Divisibility Filter: (Odd Sum - Even Sum) = 0 or 11k eliminates 3 wrong options in 2s.

Q6. Find the curved surface area (CSA) of a cone with base radius 14 cm and slant height 25 cm. [CSA = $\pi r l$]

Options: A) 1,090 cm² B) 1,100 cm² C) 1,115 cm² D) 1,120 cm²

≤5s Solution: Formula has $\pi \rightarrow$ 11-Divisibility Filter: Option B (1100) has (1+0)-(1+0) = 0. Only B is divisible by 11 in 1s. → Answer: B (1,100 cm²)

Q7. A solid metallic sphere of radius 6 cm is melted and recast into small spheres of radius 2 cm. How many small spheres are formed?

Options: A) 9 B) 18 C) 27 D) 36

≤5s Solution: $N = (R/r)^3 = (6/2)^3 = 3^3 = 27$ in 1s. → Answer: C (27)

Topic #4: Trigonometry & Heights (Expected: 1 – 4 Qs Qs | Priority: High)

Speed Target: ≤5 Sec

Master Formula: $\sin^2\theta + \cos^2\theta = 1$ | $\tan A \times \tan B = 1$ (A+B=90°) | Height = $\sqrt{a \times b}$

≤5-Second Shortcut: Angle Pinning: Tan/Cot $\rightarrow$ put $\theta = 45^\circ$; Sin/Cos $\rightarrow$ put $\theta = 90^\circ$ or 0° .

Q8. Find the value of: $\sin^2 25^\circ + \sin^2 65^\circ + \tan 35^\circ \times \tan 55^\circ$.

Options: A) 1 B) 2 C) 0 D) $\sqrt{2}$

≤5s Solution: Complementary Lock: $\sin^2 A + \sin^2 B = 1$ and $\tan A \times \tan B = 1$ when A+B=90° $\rightarrow 1 + 1 = 2$ in 1s. → Answer: B (2)

Q9. Find value of: $(1 + \tan\theta + \sec\theta)(1 + \cot\theta - \csc\theta)$.

Options: A) 0 B) 1 C) 2 D) -1

≤5s Solution: Put $\theta = 45^\circ$: $(1 + 1 + \sqrt{2})(1 + 1 - \sqrt{2}) = (2 + \sqrt{2})(2 - \sqrt{2}) = 4 - 2 = 2$ in 2s. → Answer: C (2)

Topic #5: Profit, Loss & Discount (Expected: 1 – 3 Qs Qs | Priority: High)

Speed Target: ≤5 Sec

Master Formula: $MP / CP = (100 + \text{Profit \%}) / (100 - \text{Discount \%})$ | Dishonest Profit = $[(\text{True} - \text{False}) / \text{False}] \times 100$

≤5-Second Shortcut: Single-step MP/CP ratio. Buy X Get Y Free → Discount = Free / (Bought + Free).

Q10. A merchant allows 10% discount on MP and still earns 20% profit. If CP is Rs 450, what is MP?

Options: A) Rs 540 B) Rs 600 C) Rs 650 D) Rs 700

≤5s Solution: $MP / CP = (100 + P\%) / (100 - D\%) \rightarrow MP / 450 = 120 / 90 = 4 / 3 \rightarrow MP = 450 \times (4/3) = \text{Rs } 600$ in 3s. → Answer: B (Rs 600)

Q11. A dishonest dealer claims to sell at CP but uses 800g instead of 1kg. Find his profit %.

Options: A) 20% B) 25% C) 16.66% D) 30%

≤5s Solution: Profit % = $[(1000 - 800) / 800] \times 100 = 200 / 800 = 1/4 = 25\%$ in 2s. → Answer: B (25%)

Topic #6: Mixture & Alligation (Expected: 0 – 3 Qs Qs | Priority: Medium)

Speed Target: ≤5 Sec

Master Formula: Ratio = $(C2 - \text{Mean}) : (\text{Mean} - C1)$ | Remaining Pure = Initial $\times [1 - (y/x)]^n$

≤5-Second Shortcut: Direct cross subtraction for price/concentration. Decimal multiplier 0.9^n for replacement.

Q12. In what ratio must rice at Rs 60/kg be mixed with rice at Rs 75/kg to obtain a mixture worth Rs 65/kg?

Options: A) 1 : 2 B) 2 : 1 C) 3 : 2 D) 2 : 3

≤5s Solution: Alligation Cross: $(75 - 65) : (65 - 60) = 10 : 5 = 2 : 1$ in 2s. → Answer: B (2 : 1)

Q13. From 80L pure milk, 8L is drawn out and replaced with water. Repeated 2 times. Remaining pure milk = ?

Options: A) 64.8 L B) 65.61 L C) 72 L D) 58.32 L

≤5s Solution: Pure = $80 \times (1 - 8/80)^2 = 80 \times (9/10)^2 = 80 \times 0.81 = 64.8 \text{ L}$ in 3s. → Answer: A (64.8 L)

Topic #7: Speed, Time & Distance (Expected: 0 – 2 Qs Qs | Priority: Medium)

Speed Target: ≤5 Sec

Master Formula: Distance = $[(S1 \times S2) / |S1 - S2|] \times \Delta T$ | Average Speed = $2S1S2 / (S1 + S2)$

≤5-Second Shortcut: Early + Late → Add ΔT ; Late + Late → Subtract ΔT . Always convert minutes to hours ($\div 60$).

Q14. Walking at 6 km/h arrives 15 min late; at 8 km/h arrives 5 min early. Distance to destination = ?

Options: A) 6 km B) 8 km C) 10 km D) 12 km

≤5s Solution: $D = [(S1 \times S2) / |S1 - S2|] \times \Delta T = [(6 \times 8) / 2] \times (20/60) = 24 \times (1/3) = 8 \text{ km}$ in 3s. → Answer: B (8 km)

Topic #8: Time and Work / Pipes (Expected: 1 – 2 Qs Qs | Priority: Medium)

Speed Target: ≤5 Sec

Master Formula: Total Work = LCM + (Work of person who leaves early) | Leak Time = $(A \times B) / |A - B|$

≤5-Second Shortcut: Never subtract partial work. Add missing work of early leaver to total LCM and divide by combined efficiency.

Q15. A does work in 15 days, B in 20 days. B leaves 2 days before completion. Total time to complete work = ?

Options: A) 8 days B) 9.7 days C) 10 days D) 12 days

≤5s Solution: Total work = LCM(15,20) = 60. Add B's 2 days: $60 + 2(3) = 66$. Days = $66 / (4+3) = 66/7 = 9.7 \text{ days}$ in 4s. → Answer: B (9.7 days)

Q16. Pipe fills tank in 6h; with leak fills in 8h. Time for leak alone to empty full tank = ?

Options: A) 18 hours B) 24 hours C) 28 hours D) 30 hours

≤5s Solution: Leak Time = $(A \times B) / |A - B| = (6 \times 8) / (8 - 6) = 48 / 2 = 24 \text{ hours}$ in 2s. → Answer: B (24 hours)

Topic #9: Algebra (Power Ladders & Identities) (Expected: 1 – 2 Qs Qs | Priority: Medium)

Speed Target: ≤5 Sec

Master Formula: $x^2 + 1/x^2 = k^2 - 2$ | $x^3 + 1/x^3 = k^3 - 3k$ | $a+b+c=0 \Rightarrow a^3+b^3+c^3 = 3abc$

≤5-Second Shortcut: Degrees of freedom value putting: Set $c = -(a+b)$ or set $a=1, b=1, c=-2$.

Q17. If $x + 1/x = 3$, find $x^3 + 1/x^3$.

Options: A) 18 B) 27 C) 24 D) 21

≤5s Solution: $x^3 + 1/x^3 = k^3 - 3k = 3^3 - 3(3) = 27 - 9 = 18$ in 1s. → Answer: A (18)

Q18. If $a + b + c = 0$, find $(a^2 / bc) + (b^2 / ca) + (c^2 / ab)$.

Options: A) 0 B) 1 C) 3 D) -1

≤5s Solution: Cyclic Identity: When $a+b+c=0 \rightarrow$ expression is constant = 3 in 1s. → Answer: C (3)

Topic #10: Number System & Divisibility (Expected: 1 – 2 Qs Qs | Priority: Medium)

Speed Target: ≤5 Sec

Master Formula: $72 = 8 \times 9$ | $88 = 8 \times 11$ | Fermat $a^{p-1} \equiv 1 \pmod{p}$ | Unit Digit Cyclicity mod 4

≤5-Second Shortcut: Split composite divisors into coprime pairs. Tail for 8 (last 3 digits), Head for 9 (sum of digits).

Q19. If 8-digit number 789x531y is divisible by 72 (8×9), find $(5x - 3y)$ for smallest y.

Options: A) 12 B) 15 C) 18 D) 21

≤5s Solution: Tail for 8: $31y \div 8 \rightarrow y=2$. Head for 9: Sum = $35+x \rightarrow x=1$. $5(1)-3(2) = -1 \rightarrow$ Option A in 4s. → Answer: A (12)

Q20. Find remainder when 2^{100} is divided by 101.

Options: A) 1 B) 2 C) 100 D) 0

≤5s Solution: Fermat's Little Theorem: $2^{100} \equiv 1 \pmod{101}$. Remainder = 1 in 1s. → Answer: A (1)

Topic #11: Average (AP & Replacement) (Expected: 0 – 2 Qs Qs | Priority: Low-Medium)

Speed Target: ≤5 Sec

Master Formula: AP Average = Middle Term | New Entry = New Avg + (Old Count × ΔAvg)

≤5-Second Shortcut: Deviation Method: Never calculate sum of products. Work strictly on net surplus/deficit.

Q21. The average of 7 consecutive odd numbers is 27. What is the largest of these numbers?

Options: A) 29 B) 31 C) 33 D) 35

≤5s Solution: In consecutive AP, average = middle term (4th number = 27). 5th=29, 6th=31, 7th=33 in 2s. → Answer: C (33)

Q22. Average age of 24 students is 15 years. If teacher is included, average increases by 1 year. Teacher's age = ?

Options: A) 39 years B) 40 years C) 41 years D) 42 years

≤5s Solution: Teacher's Age = New Average + (Initial Count × Increase) = $16 + (24 \times 1) = 40$ years in 2s. → Answer: B (40 years)

Topic #12: Ratio & Proportion (Expected: 0 – 2 Qs Qs | Priority: Low-Medium)

Speed Target: ≤5 Sec

Master Formula: Mean Proportional = $\sqrt{ab}$ | Third Proportional = b^2/a | Fourth Proportional = $(b \times c)/a$

≤5-Second Shortcut: Direct proportion formulas. Bag coins: multiply denomination value by ratio units.

Q23. Find the third proportional to 16 and 24.

Options: A) 32 B) 36 C) 40 D) 48

≤5s Solution: Third proportional $c = b^2 / a = 24^2 / 16 = 576 / 16 = 36$ in 2s. → Answer: B (36)

Topic #13: Compound & Simple Interest (Expected: 0 – 2 Qs Qs | Priority: Low-Medium)

Speed Target: ≤5 Sec

Master Formula: Diff_2 = $P(R/100)^2$ | Diff_3 = $P(R/100)^2 \times [(300+R)/100]$ | CI Installment $P = x/(1+R/100) + x/(1+R/100)^2$

≤5-Second Shortcut: Effective Rate Method: 2 years at 10% → SI=20%, CI=21%, Diff=1%.

Q24. Difference between CI and SI on Rs 15,000 at 8% per annum for 2 years = ?

Options: A) Rs 80 B) Rs 96 C) Rs 108 D) Rs 120

≤5s Solution: Diff_2 = $P(R/100)^2 = 15,000 \times (64 / 10000) = 1.5 \times 64 = \text{Rs } 96$ in 2s. → Answer: B (Rs 96)

Topic #14: Percentages (The AB Ladder) (Expected: 0 – 1 Qs Qs | Priority: Low)

Speed Target: ≤5 Sec

Master Formula: Price Rise +a/b → Consumption Drop -a/(a+b) | Net Change = $[x + y + xy/100] \%$

≤5-Second Shortcut: Fraction ladder converts percentage word problems to 1-second mental deductions.

Q25. If cooking oil price rises by 25%, by what % must consumption drop for constant expenditure?

Options: A) 25% B) 20% C) 16.66% D) 30%

≤5s Solution: +1/4 rise → $-1/(4+1) = -1/5 = 20\%$ drop in 1s. → Answer: B (20%)

Topic #15: Partnership (Expected: 0 – 1 Qs Qs | Priority: Low)

Speed Target: ≤5 Sec

Master Formula: Profit Share Ratio = (Capital 1 × Time 1) : (Capital 2 × Time 2)

≤5-Second Shortcut: Cancel common zeroes before multiplying. Active partner gets salary before profit split.

Q26. A invests Rs 4,000 for 12 months; B invests Rs 6,000 for 8 months. Ratio of their profit shares = ?

Options: A) 1 : 1 B) 2 : 3 C) 3 : 2 D) 4 : 3

≤5s Solution: Profit Ratio = $(4000 \times 12) : (6000 \times 8) = 48000 : 48000 = 1 : 1$ in 2s. → Answer: A (1 : 1)

Topic #16: Boat & Stream (Expected: 0 – 1 Qs Qs | Priority: Low)

Speed Target: ≤5 Sec

Master Formula: Boat Speed / Stream Speed = $(T_{\text{upstream}} + T_{\text{downstream}}) / (T_{\text{upstream}} - T_{\text{downstream}})$

≤5-Second Shortcut: Direct speed-to-time ratio whenever distance upstream equals distance downstream.

Q27. A boat takes 6h upstream and 4h downstream over same distance. Stream = 2 km/h. Boat speed in still water = ?

Options: A) 8 km/h B) 10 km/h C) 12 km/h D) 14 km/h

≤5s Solution: $B / S = (T_u + T_d) / (T_u - T_d) \rightarrow B / 2 = 10 / 2 = 5 \rightarrow B = 10 \text{ km/h}$ in 3s. → Answer: B (10 km/h)

Topic #17: Speed of Train (Expected: 0 – 1 Qs Qs | Priority: Low)

Speed Target: ≤5 Sec

Master Formula: Time = $(\text{Length}_{\text{train}} + \text{Length}_{\text{platform}}) / \text{Speed}_{\text{train}}$ | km/h to m/s = Multiply by 5/18

≤5-Second Shortcut: Single train crossing stationary object: sum lengths. Two trains crossing: sum lengths & divide by relative speed.

Q28. A 180m train travelling at 72 km/h crosses a 220m bridge. Time taken = ?

Options: A) 15s B) 20s C) 25s D) 30s

≤5s Solution: Speed = $72 \times (5/18) = 20$ m/s. Total Distance = $180 + 220 = 400$ m. Time = $400 / 20 = 20$ s in 3s. → Answer: B (20s)

SECTION 2: GENERAL INTELLIGENCE & REASONING**25 Qs | 50 Marks | Target: 24-25 Correct****Topic #1: Coding-Decoding (Expected: 2 – 4 Qs | Priority: Very High)****Speed Target: ≤5 Sec****Master Formula: Sum-27 Opposite Letters (A-Z, B-Y, C-X, D-W, E-V, F-U, G-T, H-S, I-R, J-Q, K-P, L-O, M-N)****≤5-Second Shortcut: Check first and last letter shifts. Reverse positional sum: Sum of (27 - Position).**

Q29. If 'KING' is coded as 'PRMT' and 'AZ' is 'ZA', how is 'LOVE' coded?

Options: A) OLEV B) OLVE C) OLEW D) OLVE

≤5s Solution: Sum-27 Opposites: L(12)→O(15), O(15)→L(12), V(22)→E(5), E(5)→V(22) → OLEV in 2s. → Answer: A (OLEV)

Q30. If 'CAT' is coded as 24 and 'DOG' as 26, how is 'BAT' coded?

Options: A) 20 B) 23 C) 25 D) 28

≤5s Solution: Positional Sum: BAT = 2 + 1 + 20 = 23 in 1s. → Answer: B (23)**Topic #2: Analogy (Number & Word) (Expected: 0 – 4 Qs | Priority: Very High)****Speed Target: ≤5 Sec****Master Formula: Number Patterns: $n : n^2 \pm k$, $n : n(n+1)$, $n : n^2 \pm k$ | Semantic Precision: Instrument-Quantity****≤5-Second Shortcut: Scan prime/square jumps. Semantic pairs: check exact scientific function.**

Q31. Select related number: 7 : 56 :: 9 : ?

Options: A) 72 B) 90 C) 81 D) 99

≤5s Solution: Pattern: $n : n(n+1) \rightarrow 7 : 7(8)=56 \rightarrow 9 : 9(10)=90$ in 1s. → Answer: B (90)

Q32. Thermometer : Temperature :: Hygrometer : ?

Options: A) Pressure B) Humidity C) Density D) Current

≤5s Solution: Thermometer measures temperature; Hygrometer measures humidity in 1s. → Answer: B (Humidity)**Topic #3: Syllogisms (100-50 Method) (Expected: 0 – 3 Qs | Priority: High)****Speed Target: ≤5 Sec****Master Formula: Universal (All=100/50, No=100/100) | Particular (Some=50/50, Some Not=50/100) | Pos+Pos=Pos | Neg+Neg=No Conclusion****≤5-Second Shortcut: Zero Venn Diagrams: 100 distributes to 50; 50 cannot distribute to 100. Complementary pair = Either/Or.**

Q33. Statements: All Tables are Desks. All Desks are Benches. Conclusions: I. All Tables are Benches. II. Some Benches are Tables.

Options: A) Only I follows B) Only II follows C) Both I and II follow D) Neither follows

≤5s Solution: Universal Positive + Universal Positive = Universal Positive. Both follow in 2s. → Answer: C (Both follow)

Q34. Statements: No Dog is a Cat. No Cat is a Rat. Conclusions: I. No Dog is a Rat. II. Some Dogs are Rats.

Options: A) Only I follows B) Only II follows C) Either I or II follows D) Neither follows

≤5s Solution: Negative + Negative = No definite conclusion. Complementary pair (No + Some) → Either I or II follows in 2s. → Answer: C (Either I or II)**Topic #4: Missing Number Series (Expected: 1 – 3 Qs | Priority: High)****Speed Target: ≤5 Sec****Master Formula: Difference Level 1 & 2 | Square/Cube Steps $n^2 \pm k$ | 15-Second Skip Defense****≤5-Second Shortcut: If differences do not reveal pattern in 15 seconds, flag and skip immediately.**

Q35. Find missing term: 4, 7, 12, 19, 28, ?

Options: A) 37 B) 39 C) 41 D) 43

≤5s Solution: Differences: +3, +5, +7, +9, (+11). Next = 28 + 11 = 39 in 2s. → Answer: B (39)**Topic #5: Alphanumeric & Repeating Series (Expected: 1 – 3 Qs | Priority: High)****Speed Target: ≤5 Sec****Master Formula: Block Division: Count total characters (including blanks) & divide by factor (12=3x4, 16=4x4, 20=4x5)****≤5-Second Shortcut: Look for repeating alphabetical cycles across equal partition blocks.**

Q36. Complete the series: ccbab _ caa _ bccc _ a _

Options: A) abac B) bacb C) acba D) bbac

≤5s Solution: Count total elements (16) → divide into 4 equal blocks of 4: ccbab / caab / cccb / aabb → Option A in 3s. → Answer: A**Topic #6: Blood Relation (Coded & Pointer) (Expected: 1 – 2 Qs | Priority: Medium)****Speed Target: ≤5 Sec****Master Formula: Generation Tracking (+1 Parent, 0 Sibling, -1 Child) | Gender Filter (+ Male, - Female)****≤5-Second Shortcut: Single-line generation scan. Pointer type: start reading from 'my / his / her'.**

Q37. If 'A + B' means A is father of B; 'A × B' means A is brother of B. Which shows P is uncle of Q?

Options: A) $P \times R + Q$ B) $P + R \times Q$ C) $P - R \times Q$ D) $P \times R - Q$ **≤5s Solution: Uncle = Brother of Father → $P \times R$ (brother) + Q (father) → Option A in 2s. → Answer: A ($P \times R + Q$)**

Topic #7: Classification (Odd One Out) (Expected: 1 – 2 Qs Qs | Priority: Medium)**Speed Target: ≤5 Sec****Master Formula:** Prime Numbers | Perfect Cubes / Squares | Positional Letter Differences**≤5-Second Shortcut:** Test cube/square invariants first, then prime status, then digit sums.

Q38. Find the odd one out: A) 1331, B) 2197, C) 3375, D) 4095.

Options: A) 1331 B) 2197 C) 3375 D) 4095

≤5s Solution: $1331=11^3$, $2197=13^3$, $3375=15^3$. 4095 is not a perfect cube ($4096=16^3$) → Option D in 2s. → Answer: D (4095)**Topic #8: Dice & Cube (Expected: 0 – 2 Qs Qs | Priority: Medium)****Speed Target: ≤5 Sec****Master Formula:** 2-Matching Cancellation (Third faces are opposite) | 1-Matching Clockwise Loop | Open Net Alternate Skip**≤5-Second Shortcut:** Kill the twins, the lonely wins. Alternate boxes along straight line in unfolded net are opposite.

Q39. Two views of a dice: View 1: (4, 1, 6). View 2: (4, 1, 3). What number is opposite 6?

Options: A) 2 B) 3 C) 5 D) 1

≤5s Solution: Cancel common faces 4 and 1 on both views → 6 is opposite 3 in 1s. → Answer: B (3)**Topic #9: Mathematical Operator (Expected: 0 – 2 Qs Qs | Priority: Medium)****Speed Target: ≤5 Sec****Master Formula:** BODMAS Strict Order: Brackets → Orders → Division → Multiplication → Addition → Subtraction**≤5-Second Shortcut:** Division Feasibility Test: Check if division yields an integer; eliminates invalid sign interchanges.Q40. If '+' means 'x', '-' means '+', 'x' means '+', '+' means '-', find value of: $20 + 5 \times 10 \div 6 - 2$.

Options: A) 107 B) 112 C) 100 D) 95

≤5s Solution: Substitute: $20 \times 5 + 10 - 6 \div 2 = 100 + 10 - 3 = 107$ in 3s. → Answer: A (107)**Topic #10: Arithmetic Reasoning (Heads & Legs) (Expected: 0 – 2 Qs Qs | Priority: Medium)****Speed Target: ≤5 Sec****Master Formula:** 4-Legged Animals = (Total Legs / 2) - Total Heads | 2-Legged Animals = Total Heads - 4-Legged**≤5-Second Shortcut:** All-2-legged assumption: Extra legs divided by 2 gives 4-legged animal count instantly.

Q41. A farmer has chickens and goats. Total heads = 40, total legs = 110. How many goats are there?

Options: A) 15 B) 20 C) 25 D) 30

≤5s Solution: All 40 chickens = 80 legs. Extra legs = $110 - 80 = 30$. Each goat has 2 extra legs → Goats = $30/2 = 15$ in 2s. → Answer: A (15)**Topic #11: Seating Arrangement (Expected: 0 – 1 Qs Qs | Priority: Low-Medium)****Speed Target: ≤5 Sec****Master Formula:** Linear Row (Facing North: Right is East, Left is West) | Circular (Facing Center: Right is Anti-Clockwise)**≤5-Second Shortcut:** Anchor the definite center/end person first and fill adjacent relative positions.

Q42. Five friends A, B, C, D, E sit in a row facing North. C sits in the middle. A is left of B but right of C. Who is on immediate right of C?

Options: A) A B) B C) D D) E

≤5s Solution: A is right of C and left of B → Array: __ C A B → Immediate right of C is A in 2s. → Answer: A (A)**Topic #12: Paper Cutting & Folding (Expected: 0 – 1 Qs Qs | Priority: Low-Medium)****Speed Target: ≤5 Sec****Master Formula:** Number of Punches = $2^{(\text{folds})} \times \text{Original Punches}$ | Symmetry Reflection across Fold Lines**≤5-Second Shortcut:** Trace the final punch mark; it must appear in the exact same quadrant in the answer figure.

Q43. A circular sheet is folded twice into a quadrant and punched at corner. Unfolded sheet shows:

Options: A) 2 punches B) 4 symmetrical punches forming diamond C) 8 punches D) 1 punch

≤5s Solution: 2 folds = 4 layers → 4 symmetrical punches reflecting across both axes → Option B in 1s. → Answer: B**Topic #13: Logical Venn Diagram (Expected: 0 – 1 Qs Qs | Priority: Low-Medium)****Speed Target: ≤5 Sec****Master Formula:** Subset Inclusion (Concentric) | Disjoint Sets (Separate Circles) | Overlapping Attributes (Intersecting)**≤5-Second Shortcut:** Identify universal subset hierarchies (A is always B).

Q44. Which diagram represents: 'Reptiles, Lizards, Animals'?

Options: A) Three intersecting circles B) Three concentric circles C) Two disjoint in one big D) Three separate

≤5s Solution: Lizards $\subset$ Reptiles $\subset$ Animals → Three Concentric Circles in 1s. → Answer: B (Three concentric)**Topic #14: Rank and Order (Expected: 0 – 1 Qs Qs | Priority: Low)****Speed Target: ≤5 Sec****Master Formula:** Total T = (Rank Left + Rank Right) - 1 | Person Swapping Total = New Rank A + Old Rank B - 1**≤5-Second Shortcut:** Always subtract 1 to eliminate double-counting of the anchor person.

Q45. Rahul is 12th from left and 18th from right in a row. Total students = ?

Options: A) 29 B) 30 C) 31 D) 28

≤5s Solution: Total = Left + Right - 1 = 12 + 18 - 1 = 29 in 1s. → Answer: A (29)

Topic #15: Mirror & Water Image (Expected: 0 – 1 Qs Qs | Priority: Low) Speed Target: ≤5 Sec

Master Formula: Mirror Image: Lateral Inversion (Left ↔ Right) | Water Image: Vertical Inversion (Top ↔ Bottom)

≤5-Second Shortcut: Asymmetrical letter check: E, F, G, J, L, P, Q, R, S, Z flip laterally in mirror.

Q46. Select mirror image of 'WHITE' with mirror placed on right.

Options: A) ETIHW B) Inverted letters starting with mirror E C) WHITE D) Inverted W

≤5s Solution: Right mirror reverses word order AND laterally inverts each letter → Option B in 1s. → Answer: B

SECTION 3: ENGLISH COMPREHENSION**25 Qs | 50 Marks | Target: 23-24 Correct****Topic #1: Cloze Test (Grammar & Context) (Expected: 3 – 5 Qs | Priority: Very High)****Speed Target: ≤5 Sec****Master Formula: Tone Alignment (Positive/Negative sentiment matching) | Fixed Preposition Collocations****≤5-Second Shortcut: Read full sentence before looking at options. Preposition collocations eliminate 3 wrong choices.****Q47. Fill blank: 'Regular cardiovascular exercise is highly conducive _____ good health.'****Options:** A) for B) to C) with D) in**≤5s Solution: Fixed Preposition Collocation: Conducive TO in 1s. → Answer: B (to)****Topic #2: Direct-Indirect & Active-Passive (Expected: 2 – 4 Qs | Priority: Very High)****Speed Target: ≤5 Sec****Master Formula: Voice Tense-Immunity (Tense NEVER changes) | Narration Said-Backshift (Will → Would, Past → Past Perfect)****≤5-Second Shortcut: Scan verbs only: Voice changes form not tense; Narration backshifts modal auxiliary.****Q48. Select passive: 'The chef prepared a sumptuous meal for the guests.'****Options:** A) A sumptuous meal is prepared.
B) A sumptuous meal was prepared by the chef for the guests.
C) A sumptuous meal had been prepared.
D) A sumptuous meal has been prepared.**≤5s Solution: Tense Immunity: Past Simple (V2: prepared) MUST become was/were + V3 → Option B in 1s. → Answer: B****Q49. Convert to indirect: Rahul said, 'I will call you tomorrow.'****Options:** A) Rahul said that he will call me.
B) Rahul said that he would call me the next day.
C) Rahul said that he should call me.
D) Rahul said that he can call me.**≤5s Solution: Said-Backshift: Said + Will MUST become WOULD; tomorrow becomes the next day → Option B in 1s. → Answer: B****Topic #3: Vocabulary (Synonyms & Antonyms) (Expected: 2 – 4 Qs | Priority: Very High)****Speed Target: ≤5 Sec****Master Formula: Etymology Root Words (Mal=Bad, Bene=Good, Loqu=Speak, Chron=Time, Phil=Love, Path=Feeling)****≤5-Second Shortcut: Deduce word polarity from Latin/Greek prefixes before reading full definitions.****Q50. What is the meaning of 'Loquacious'?****Options:** A) Silent B) Highly talkative C) Wealthy D) Cruel**≤5s Solution: Root Word LOQU = Speak → Loquacious = Talkative in 1s. → Answer: B (Highly talkative)****Topic #4: Fill in the Blanks (Expected: 1 – 4 Qs | Priority: High)****Speed Target: ≤5 Sec****Master Formula: Subject-Verb Agreement | Collective Noun Division (United=Singular, Divided=Plural)****≤5-Second Shortcut: Check if collective noun acts as single unit or divided individuals.****Q51. Select word: 'The jury _____ divided in their opinions regarding the verdict.'****Options:** A) was B) were C) is D) has been**≤5s Solution: Collective noun acting as individuals (divided in opinions) takes PLURAL verb: 'were' in 2s. → Answer: B (were)****Topic #5: Spelling Test (Expected: 1 – 4 Qs | Priority: High)****Speed Target: ≤5 Sec****Master Formula: Double Consonant Patterns (cc/mm in Accommodate, rr/ss in Embarrass, cc/ss in Occasion)****≤5-Second Shortcut: I before E except after C (Receive, Deceive vs Believe, Achieve).****Q52. Identify correctly spelt word: A) Accomodate, B) Accommodate, C) Accommodate, D) Acomodate.****Options:** A) Accomodate B) Accommodate C) Accommodate D) Acomodate**≤5s Solution: Double-C and Double-M Rule: Ac-com-mo-date → Option B in 1s. → Answer: B (Accommodate)****Topic #6: Para Jumbles (PQRS) (Expected: 0 – 3 Qs | Priority: Medium)****Speed Target: ≤5 Sec****Master Formula: Noun-Pronoun Mandatory Pairs | Acronym Precedence (Full Name → Acronym) | Conjunction Closers****≤5-Second Shortcut: Find mandatory opener (Noun) and eliminate options that begin with pronouns/conjunctions.****Q53. Arrange: P: He formulated laws of motion. Q: Isaac Newton was born in 1643. R: This changed science. S: He published Principia.****Options:** A) PQRS B) QPSR C) RQPS D) SPRQ**≤5s Solution: Noun introduces subject first (Q: Newton), Pronoun follows (P: He) → Q MUST precede P → Option B in 2s. → Answer: B (QPSR)****Topic #7: Idioms & Phrases (Expected: 1 – 2 Qs | Priority: Medium)****Speed Target: ≤5 Sec****Master Formula: Non-Literal Interpretation (Literal physical meaning is 100% false distractor)**

≤5-Second Shortcut: Link idiom metaphor to psychological/social human behavior.

Q54. Meaning of the idiom: 'To bury the hatchet'?

Options: A) To dig a hole B) To make peace C) To start war D) To lose a tool

≤5s Solution: Non-literal interpretation: To bury the hatchet = To end a conflict and make peace in 1s. → Answer: B (To make peace)

Topic #8: One Word Substitution (Expected: 0 – 2 Qs Qs | Priority: Medium)

Speed Target: ≤5 Sec

Master Formula: Category Classification: Incorrigible (cannot be reformed), Infallible (cannot make mistakes)

≤5-Second Shortcut: Root matching: -cide (killing), -phobia (fear), -phil (lover), -cracy (rule).

Q55. A person who cannot be corrected or reformed:

Options: A) Infallible B) Incorrigible C) Invincible D) Indelible

≤5s Solution: Cannot be corrected = Incorrigible in 1s. → Answer: B (Incorrigible)

Topic #9: Sentence Completion & Error Spotting (Expected: 0 – 1 Qs Qs | Priority: Low-Medium)

Speed Target: ≤5 Sec

Master Formula: Preposition Fluff Cloak | Subjunctive 'Were' | Correlative Inversion (Hardly...when, No sooner...than)

≤5-Second Shortcut: Blindfold prepositional phrases between subject and verb to verify true singular/plural agreement.

Q56. Find error: 'The price of these new smartphones have skyrocketed in recent months.'

Options: A) The price of B) these new smartphones C) have skyrocketed D) in recent months

≤5s Solution: Preposition Fluff Cloak: Blindfold 'of these new smartphones'. Singular 'price' requires 'HAS skyrocketed' → Error C in 2s. → Answer: C (have skyrocketed)

Q57. Find error: 'If I was the Prime Minister, I would eliminate corruption.'

Options: A) If I was B) the Prime Minister C) I would eliminate D) corruption

≤5s Solution: Subjunctive Were: Hypothetical condition ALWAYS takes 'WERE' → Error in part A in 1s. → Answer: A (If I was)

SECTION 4: GENERAL AWARENESS**25 Qs | 50 Marks | Target: 19-21 Correct****Topic #1: Current Affairs (6-8 Months) (Expected: 2 – 5 Qs Qs | Priority: Very High)****Speed Target: ≤5 Sec****Master Formula: National & International Events | Defense Exercises | Major Appointments | Sports Awards****≤5-Second Shortcut: Focus on last 6 months prior to exam date: Appointments, Grand Slam winners, Padma Awards.****Q58. Who was appointed as the Chief of the Naval Staff of India in recent defense appointments?****Options:** A) Admiral R. Hari Kumar B) Admiral Dinesh K. Tripathi C) General Anil Chauhan D) Air Chief Marshal VR Chaudhari**≤5s Solution: Recent appointment: Admiral Dinesh K. Tripathi in 1s. → Answer: B****Topic #2: History (Ancient, Medieval, Modern) (Expected: 1 – 5 Qs Qs | Priority: Very High)****Speed Target: ≤5 Sec****Master Formula: Delhi Sultanate S-K-T-S-L | Mughal B-H-A-J-S-A | Major INC Sessions & Acts (1909, 1919, 1935)****≤5-Second Shortcut: Anchor key years: Champaran 1917, Non-Cooperation 1920, Civil Disobedience 1930, Quit India 1942.****Q59. Which dynasty ruled Delhi Sultanate immediately after the Khilji dynasty?****Options:** A) Slave Dynasty B) Tughlaq Dynasty C) Sayyid Dynasty D) Lodi Dynasty**≤5s Solution: Mnemonic S-K-T-S-L: Slave → Khilji → Tughlaq → Sayyid → Lodi → Option B in 1s. → Answer: B (Tughlaq Dynasty)****Q60. In which session did Mahatma Gandhi preside over the Indian National Congress?****Options:** A) 1916 Lucknow B) 1924 Belgaum C) 1929 Lahore D) 1931 Karachi**≤5s Solution: Gandhi presided over ONLY ONE INC session: 1924 Belgaum in 1s. → Answer: B (1924 Belgaum)****Topic #3: Economy (Macro & Banking) (Expected: 1 – 5 Qs Qs | Priority: Very High)****Speed Target: ≤5 Sec****Master Formula: Repo Rate (RBI → Banks) | Reverse Repo (Banks → RBI) | Cash Reserve Ratio (CRR) | Five-Year Plans****≤5-Second Shortcut: 1st Plan (1951-56: Harrod-Domar), 2nd Plan (1956-61: Mahalanobis heavy industry).****Q61. The interest rate at which the Reserve Bank of India lends short-term money to commercial banks is called:****Options:** A) Reverse Repo Rate B) Repo Rate C) Bank Rate D) CRR**≤5s Solution: RBI lends to commercial banks = Repo Rate in 1s. → Answer: B (Repo Rate)****Topic #4: Indian Polity (Center-State Rules) (Expected: 1 – 3 Qs Qs | Priority: High)****Speed Target: ≤5 Sec****Master Formula: Center-to-State +89 Rule (Articles 72-111) | Center-to-State +90 Rule (Articles 112-124)****≤5-Second Shortcut: Convert central constitutional provisions to state provisions instantly without memorizing 395 articles.****Q62. If Article 72 gives President pardoning power, which Article gives Governor pardoning power?****Options:** A) Article 153 B) Article 161 C) Article 165 D) Article 167**≤5s Solution: Center Article + 89 = State Article: 72 + 89 = Article 161 in 1s. → Answer: B (Article 161)****Q63. If Article 112 is Union Budget, which Article is State Budget?****Options:** A) Article 200 B) Article 202 C) Article 204 D) Article 206**≤5s Solution: Center Article + 90 = State Article: 112 + 90 = Article 202 in 1s. → Answer: B (Article 202)****Topic #5: Static GK (Dance Maestros & Heritage) (Expected: 2 – 3 Qs Qs | Priority: High)****Speed Target: ≤5 Sec****Master Formula: Surname Mappings: Maharaj = Kathak | Reddy = Kuchipudi | Mohapatra = Odissi****≤5-Second Shortcut: Match classical dance maestros by surname patterns; match UNESCO sites by state anchors.****Q64. Padma Vibhushan Birju Maharaj is associated with which Indian classical dance form?****Options:** A) Kathakali B) Kathak C) Bharatanatyam D) Odissi**≤5s Solution: Surname Rule: 'Maharaj' = KATHAK (Uttar Pradesh) in 1s. → Answer: B (Kathak)****Topic #6: Geography (Rivers & Passes) (Expected: 1 – 2 Qs Qs | Priority: Medium)****Speed Target: ≤5 Sec****Master Formula: West-Flowing Rivers: Narmada, Tapi, Mahi, Sabarmati (Arabian Sea, Estuaries, No Delta)****≤5-Second Shortcut: Passes: Nathu La (Sikkim), Zoji La (Ladakh), Shipki La (Himachal), Lipulekh (Uttarakhand).****Q65. Which river flows westwards into the Arabian Sea and forms an estuary?****Options:** A) Godavari B) Narmada C) Krishna D) Mahanadi**≤5s Solution: West-Flowing Rift Valley Rivers = Narmada & Tapi in 1s. → Answer: B (Narmada)****Topic #7: Everyday Chemistry (Expected: 1 – 2 Qs Qs | Priority: Medium)****Speed Target: ≤5 Sec****Master Formula: Baking Soda = NaHCO₃ | Washing Soda = Na₂CO₃·10H₂O | Bleaching Powder = CaOCl₂ | Plaster of Paris = CaSO₄·1/2H₂O**

≤5-Second Shortcut: Common chemical names and formulas recurring in TCS exams.

Q66. What is the chemical name of 'Baking Soda'?

Options: A) Sodium Carbonate B) Sodium Hydrogen Carbonate C) Calcium Carbonate D) Sodium Hydroxide

≤5s Solution: Baking Soda = Sodium Bicarbonate (NaHCO_3) in 1s. → Answer: B (Sodium Hydrogen Carbonate)

Topic #8: Biology (Anatomical Roots & Vitamins) (Expected: 0 – 2 Qs Qs | Priority: Low-Medium)

Speed Target: ≤5 Sec

Master Formula: Medical Prefixes: Nephro (Kidney), Hepato (Liver), Pneumo (Lungs), Cardio (Heart), Osteo (Bone)

≤5-Second Shortcut: Vitamins: A (Retinol), B1 (Thiamine), C (Ascorbic Acid), D (Calciferol), K (Phylloquinone).

Q67. Which organ is affected in 'Nephritis'?

Options: A) Liver B) Kidney C) Lungs D) Heart

≤5s Solution: Medical Root Nephro = Kidney → Nephritis = Kidney inflammation in 1s. → Answer: B (Kidney)

Q68. What is the chemical name of Vitamin B1?

Options: A) Riboflavin B) Thiamine C) Niacin D) Pyridoxine

≤5s Solution: Vitamin B1 = Thiamine in 1s. → Answer: B (Thiamine)

Topic #9: Physics (SI Units & Optics) (Expected: 0 – 2 Qs Qs | Priority: Low)

Speed Target: ≤5 Sec

Master Formula: SI Units: Force (Newton), Work/Energy (Joule), Power (Watt), Resistance (Ohm), Frequency (Hertz)

≤5-Second Shortcut: Optics: Concave Mirror (Real/Inverted, shaving), Convex Mirror (Virtual/Erect, rear-view).

Q69. What is the SI unit of Electric Potential Difference?

Options: A) Ampere B) Volt C) Ohm D) Watt

≤5s Solution: Potential Difference = Volt in 1s. → Answer: B (Volt)